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Porter Stemmer Algorithm: A Text Preprocessing Technique in NLP

DSA0307 - NATURAL LANGUAGE PROCESSING

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Engineer to Excel

Introduction

- Porter Stemmer is a word stemming algorithm used in NLP.
- It converts words into their root/stem form.
- Developed by Martin Porter in 1980.
- Used for text preprocessing in NLP applications.

Example:

Playing → Play

Connected → Connect

Engineer to Excel

Need and Importance

- Computers treat different word forms as separate words.
- Stemming combines similar words into a single form.
- Reduces the size of vocabulary.
- Improves the performance of NLP models.

Used in:

❖ Search engines

❖ Text classification

Working Mechanism

- It follows a series of suffix removal rules.

Main steps:

- Remove plural forms
- Remove past tense endings
- Remove continuous forms
- Remove common suffixes
- Perform final word cleanup

Advantages:

- Simple and fast algorithm.
- Reduces text complexity.
- Easy to implement.

Limitations:

- May generate incorrect root words.
- Does not understand word meaning.

Thank You!

For your attention

